# Applying Newton’s Laws

## Building Models to Describe our World

### Motion, Derivations and Modelling Situations

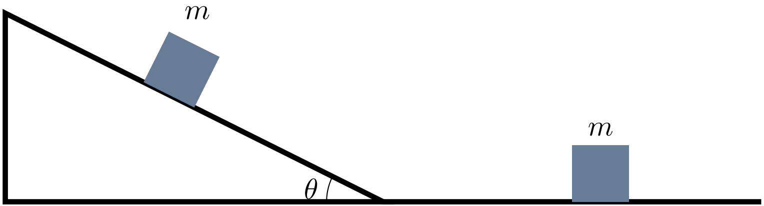
# Outline

# Types of Motion

Applying Newton’s Laws

# Linear motion

- We can loosely define "linear motion" as motion along a straight line, where the velocity does not change direction.
- The net force (and acceleration) is co-linear with the velocity.
- We can break up many types of motion into segments where the motion is linear along the segments.




*A block slides down an incline and then a flat section. Both segments are linear motion.*

# Constant net force

- When the **sum of the forces** (aka the net force) on an object is constant (in magnitude and direction), and the mass of the object is constant, the acceleration of that object is constant:

$$\sum \vec{F} = \vec{F}^{net} = m\vec{a}$$

- When the acceleration is constant:

$$\vec{v}(t) = \vec{v}_0 + \vec{a}t = \vec{v}_0 + \frac{\vec{F}^{net}}{m}t$$

$$\vec{r}(t) = \vec{r}_0 + \vec{v}_0 t + \frac{1}{2}\vec{a}t^2 = \vec{r}_0 + \vec{v}_0 t + \frac{1}{2}\frac{\vec{F}^{net}}{m}t^2$$

- This is a vector equation, so true by component, for example, the $x$ component:

$$v_x(t) = v_{0x} + \frac{F_x}{m}t \quad x(t) = x_0 + v_{0x}t + \frac{1}{2}\frac{F_x}{m}t^2$$

# Varying net force

- The net force on an object can change magnitude and direction:
  - As a function of time.
  - As a function of position.
  - As a function of velocity.
- Then, the acceleration is not constant, and we have to start with:

$$\frac{d\vec{v}}{dt} = \vec{a} = \frac{\vec{F}^{net}}{m}$$
$$\frac{d\vec{r}}{dt} = \vec{v}$$

and integrate...

# Linear motion, force varies with time

- If the magnitude of the force changes with time:

$$\vec{a}(t) = \frac{\vec{F}^{net}(t)}{m}$$

- The kinematics equations are “straightforward”:

$$\frac{d\vec{v}}{dt} = \frac{1}{m}\vec{F}^{net}(t)$$
$$\therefore \vec{v}(t) = \vec{v}_0 + \frac{1}{m}\int_0^t \vec{F}^{net}(t)dt$$
$$\therefore \vec{r}(t) = \vec{r}_0 + \int_0^t \vec{v}(t)dt$$

just some integrals that you can look up if you know the function $\vec{F}^{net}(t)$...

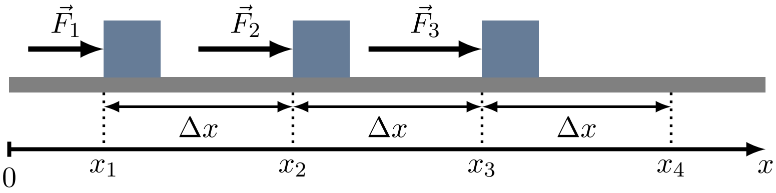
# Linear motion, force varies with position

- If the magnitude of the force changes with position of the object:

$$a_x(x) = \frac{F_x^{net}(x)}{m}$$

- We cannot “simply integrate” the equations that define acceleration from velocity and velocity from position:

$$\frac{dv_x}{dt} = a_x(x) \rightarrow v_x(t) = v_{0x} + \int_0^t a_x(x)dt$$




*A block slides along $x$ with a force that changes depending on position.*

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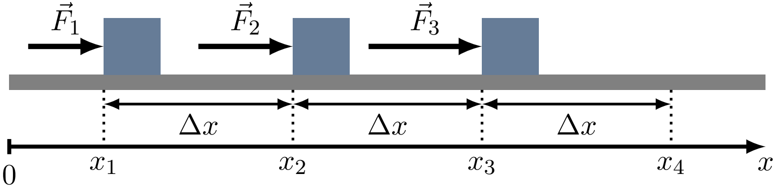


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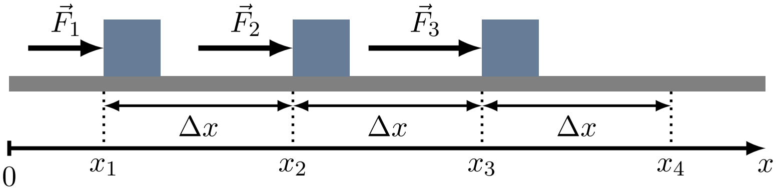
# Breaking it up

- When things vary continuously, we can break up the motion into small segments where things are constant (in this case, things = force, acceleration).

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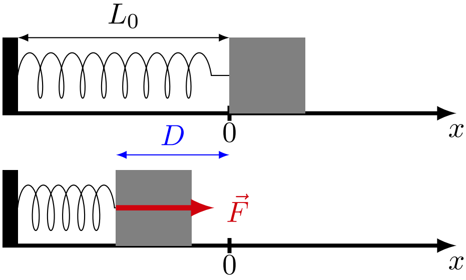
# Applying this to a block on a spring

- A block is pushed against a spring, compressing it by a distance $D$.
- The force on the block (exerted by the spring) changes with position.

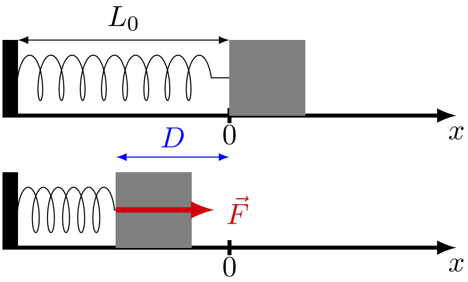



*Compressing a spring by a distance $D$.*

# Applying this to a block on a spring

- A block is pushed against a spring, compressing it by a distance $D$.
- The force on the block (exerted by the spring) changes with position.
- The velocity (squared) of the block as a function of position is given by:

$$v^2(x) = v_0^2 + \frac{2}{m}\int F(x)dx$$

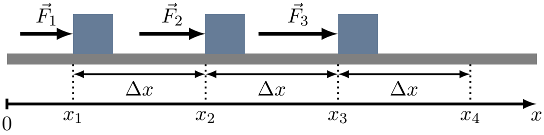



*Compressing a spring by a distance $D$.*

# Force varying with position (recap)

- We can find the speed squared by applying the kinematic equation multiple times:

$$v_1^2 = v_0^2 + 2\frac{F_1}{m}\Delta x$$
$$v_n^2 = v_0^2 + \frac{2}{m}\sum F_i \Delta x$$

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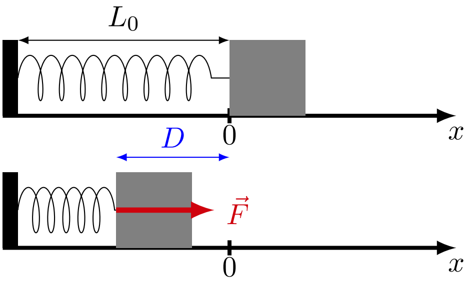
# Force varying with position (recap)

- A block is pushed against a spring, compressing it by a distance $D$. The force on the block (exerted by the spring) changes with position:

$$F(x) = -kx$$

- Having released the block at rest ($v_0 = 0$) from $x = -D$, its speed when it reaches $x = 0$ is:

$$v^2(x=0) = -\frac{2}{m}\int_{-D}^{0} kx\,dx = -\frac{2}{m}\left[\frac{1}{2}kx^2\right]_{-D}^{0}$$
$$= \frac{k}{m}D^2$$




*Compressing a spring by a distance $D$.*

# Mathmatical Applications

## Applying Newton’s Laws

# Mathematically, more formally

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- We can thus write

$$dv = a(x)dt = \frac{a(x)}{v}dx$$

# Mathematically, more formally

- Think of the Chain Rule for the same result:

$$\frac{dv}{dx} = \frac{dv}{dt}\frac{dt}{dx} = a\frac{1}{v}$$
$$\therefore dv = \frac{a}{v}dx$$

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- We can "add together" things on the right and the left, and they must still be equal:

$$\int_{v_0}^{v(x_1)} vdv = \int_{x_0}^{x_1} a(x)dx$$
$$\frac{1}{2}v^2(x) - \frac{1}{2}v_0^2 = \int_{x_0}^{x_1} a(x)dx$$
$$\therefore v^2(x) = v_0^2 + 2\int_{x_0}^{x_1} a(x)dx = v_0^2 + \frac{2}{m}\int_{x_0}^{x_1} F(x)dx$$

# Mathematically, more formally

- With constant acceleration, this just gives the kinematics equation:

$$\begin{aligned} v^2(x_1) &= v_0^2 + 2\int_{x_0}^{x_1} a(x)dx \\ &= v_0^2 + 2a(x_1 - x_0) \end{aligned}$$

- This is also where kinetic energy comes from (next week!)

$$\begin{aligned} v^2(x_1) &= v_0^2 + \frac{2}{m}\int_{x_0}^{x} F(x)dx \\ \therefore \frac{1}{2}mv^2(x_1) - \frac{1}{2}mv_0^2 &= \int_{x_0}^{x} F(x)dx \end{aligned}$$

Final kinetic energy minus initial kinetic energy is equal to Work done...

# Differential equations

- This is a differential equation:

$$v dv = a(x) dx$$

- We could write it as:

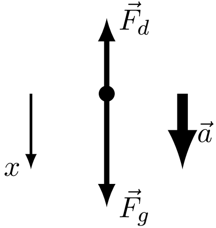
$$\frac{dv}{dx} = \frac{a(x)}{v}$$

- The derivative of a function $\frac{dv}{dx}$ depends on the function ($v(x)$) itself.
- It is a **separable differential** equation, because we can separate all that depends on $v$ and $x$ onto separate sides of the equation (as in the first line). This allows us to find $v(x)$ by integrating both sides, as in the last slides.
- Separable = usually easy to solve.
- At this point, **you're not expected to solve a differential equation**, but you should understand the concept and how we're solving a separable equation.

# Force varying with velocity (e.g. drag)

- The force of drag (aka air resistance) depends on velocity.
- Ultimately, it depends on the actual situation, sometimes it depends on $v$, sometimes on $v^2$.
- The force also depends on the medium through which the object moves, and the size of the object.
- Can model this as:

$$\vec{F}_d = -b\vec{v}$$
$$\vec{F}_d = -bv^2\hat{v}$$

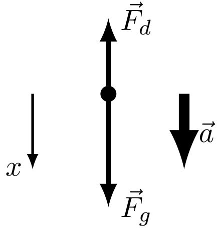



*A model for drag.*

# Example: Drag linear with velocity

- We assume drag is linear (and write the $x$ component):

$$F_d = -bv$$




*A model for drag.*

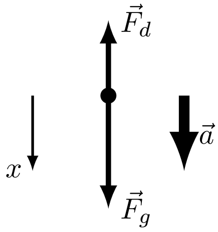
# Example: Drag linear with velocity

- We assume drag is linear (and write the $x$ component):

$$F_d = -bv$$

- Writing Newton's Second Law (which is a differential equation):

$$\sum F - x = mg - bv = ma$$
$$mg - bv = m\frac{dv}{dt}$$

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$$\sum F - x = mg - bv = ma$$

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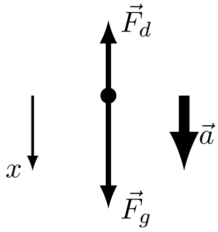
- It's separable:

$$\frac{dv}{mg - bv} = -\frac{b}{m}dt$$

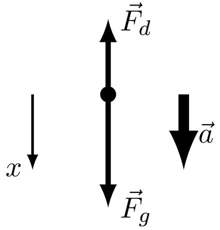



*A model for drag.*

# Example: Drag linear with velocity

- So we can integrate both sides:

$$\int_0^{v(t_1)} \frac{dv}{mg - bv} dv = -\int_0^{t_1} \frac{b}{m} dt$$

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- So we can integrate both sides:

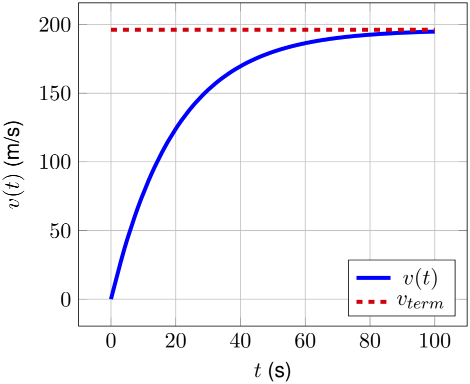
$$\int_0^{v(t_1)} \frac{dv}{mg - bv} dv = -\int_0^{t_1} \frac{b}{m} dt$$

- to find (after some math):

$$v(t_1) = \frac{mg}{b}\left(1 - e^{-\frac{b}{m}t_1}\right)$$

The speed increases asymptotically to the "terminal velocity". In the limit $t \to \infty$, we have:

$$v(t \to \infty) = \frac{mg}{b}$$




*Speed as a function of time for an object that started from rest and falls with drag. The object has a mass of* $m = 10$ kg *and the coefficient of drag is given by* $b = 0.5$ N · s/m.

# Building Models Using Newton's Laws

## Applying Newton's Laws

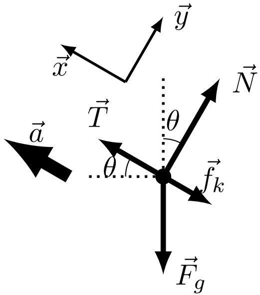
# Reminder: Applying Newton's Laws

Strongly recommend the following procedure for model building:

1. **Make a very good Free Body Diagram** for each mass that shows:
   1.1 All the forces ("what could be pushing or pulling on the mass?").
   1.2 The acceleration vector.
   1.3 Relevant angles.
   1.4 Your choice of coordinate system (make $+x$ parallel to acceleration).
2. Read off the free-body diagram(s) to write Newton's Second Law, you get one equation per dimension:

$$\sum F_x = ma \quad \sum F_y = 0$$

   (if you chose x-axis parallel to acceleration, not always possible).
3. You now have a series of equations, just do some math to solve for any variables you need.




*Reading off your very good free body diagram*

Reading off the diagram to write Newton's Second Law:

$$\begin{aligned} \sum F_x : \quad & T - f_k - F_g \sin\theta = ma \\ \sum F_y : \quad & N - F_g \cos\theta = 0 \end{aligned}$$

# Newton’s Laws and Circular motion

- In circular motion (motion along a circle):
  - The velocity is always tangent to the circle.
  - **There is always a centripetal component of acceleration** (radial acceleration, perpendicular to the velocity, responsible for changing the direction of $\vec{v}$).
  - There can be a tangential component of the acceleration, if the object changes speed (**non-uniform** circular motion).
  - The angular quantities ($\theta, \omega, \alpha$) are related to the tangential quantities (e.g. $v = \omega R$, $a_T = \alpha R$ ).
- The sum of the forces must be parallel to the total acceleration:

$$\sum \vec{F} = m\vec{a}$$

- Can break this up into radial and tangential components (a good choice of $xy$):

$$\sum F_R = ma_R = m\frac{v^2}{R} = m\omega^2 R \quad \text{(Radial component, never zero)}$$

$$\sum F_T = ma_T = m\alpha R \quad \text{(Tangential component, zero if uniform circ. mot.)}$$

# Two masses on accelerating turntable

- Both tangential and radial acceleration, only force of static friction can result in acceleration.
- Both masses have the same angular velocity ($\omega$) and angular acceleration ($\alpha$).
- Masses stay while $f_s \leq \mu_s N = \mu_s mg$.
- Tangential (linear) acceleration (constant):

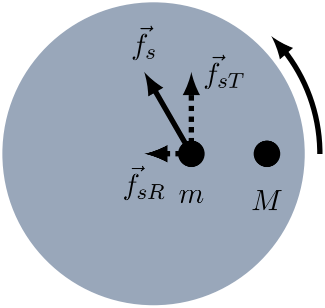
$$ma_T = m\alpha r = f_{sT}$$

- Radial (linear) acceleration (increasing):

$$ma_R = m\omega^2 r = f_{sR}$$

- Magnitude of the force of friction:

$$f_s = \sqrt{f_{sT}^2 + f_{sR}^2} = mr\sqrt{\alpha^2 + \omega^2} \leq mg$$
$$\therefore r\sqrt{\alpha^2 + \omega^2} \leq g$$

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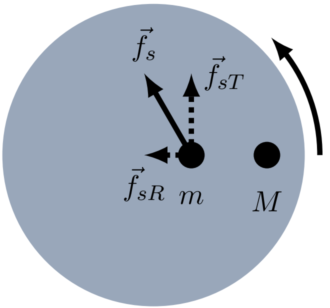



*Two masses on an accelerating turntable.*

Since $\alpha$ and $\omega$ are the same for both masses, the one with the larger value of $r$ will fall of first, regardless of mass!

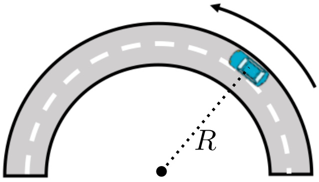
# Modeling *uniform* circular motion

- Uniform means constant speed (no tangential acceleration).
- The **net acceleration** is towards the centre of the circle:

$$a_C = a_R = m\frac{v^2}{R}$$

- The **sum of the forces** is directed towards the centre of the circle:

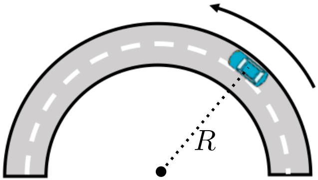
$$\sum F_R = m\frac{v^2}{R}$$
$$\sum F_T = 0$$




*Top view of car going around a corner*

# Banked curves

- By banking the road, one can change the direction of the normal force so that it, too, can have a component towards the centre of the circle

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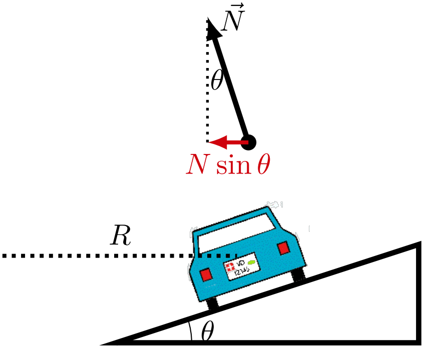
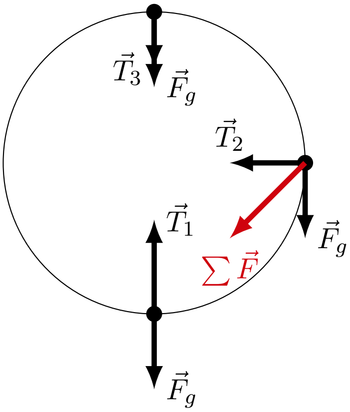



*Side view of car going around a banked curve.*

# Motion in a vertical circle

- What you did as an experiment in RA5.
- A ball swinging around a rope, with a force of tension.
- A roller coaster, with a normal force instead of tension.

*It can never be uniform circular motion, there will always be a tangential component to acceleration from gravity (except at the top and bottom).*




*A mass on a string going around in a vertical circle. The sum of the forces only points to the centre of the circle at the top and bottom.*